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ASSIGNMENT-1

Part 1. Probability

1.1 2 coins → tossed simultaneously H\T

a) $S = \{HH, HT, TH, TT\}$

b) $F = \{\phi, \{HH\}, \{HT\}, \{TH\}, \{TT\},$

2 elmts, $\{HH, HT\}, \{HH, TH\}, \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \{TH, TT\}$

3 elmt → $\{HH, HT, TH\}, \{HH, HT, TT\}, \{HH, TH, TT\}, \{HT, TH, TT\}$

4 elmts → $\{HH, HT, TH, TT\}$

Total 16 elmnts.

c) $P(\{HH\}) = P(\{HT\}) = P(\{TH\}) = P(\{TT\}) = P$

$\Rightarrow \sum P(E) = 1 \ (E \in S)$

$\Rightarrow 4P = 1$

$P = 1/4$

(i) probability $= 1/4$

(ii). atleast one head → HH, HT, TH

$P(\{HH, HT, TH\}) = 3/4 = P(\{HH\}) + P(\{HT\}) + P(\{TH\})$

(iii) exactly one head occurs in → HT, TH

$P(\{HT, TH\}) = P(\{HT\}) + P(\{TH\})$

$= 2/4$

Part 2 Discrete Random Variables

2.1. participant shown 50 words ⇒ $n = 50$

no. of correctly recognised words ⇒ $k = 45$

probability of being correct ⇒ $p = 0.9$

given $f(k,n,p) = \frac{n!}{k!(n-k)!} p^k (1-p)^{(n-k)}$

$= \frac{50!}{45!\,5!} (0.9)^{45} (0.1)^5$

$= 2118760 \times 0.008727.. \times 10^{-5}$

$= 0.1849$

$P(X=45) = f(k,n,p) = 0.1849$

2.2. 10 road accidents on average!! in a single day.

probability of k no. of road accidents in a day

⇓

$f(k,\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$ ($\lambda = 10$ given)

a) zero road accidents in a day.

$k = 0$

$f(0,10) = \frac{10^0 e^{-10}}{0!} = e^{-10} = 4.54 \times 10^{-5}$

$= 0.00004540$

b). $P(7 < X < 10)$

$= P(X=8) + P(X=9)$

$f(8,10) = \frac{10^8 \times e^{-10}}{8!} = \frac{1}{40320} \times 10^8 \times 4.54 \times 10^{-5}$

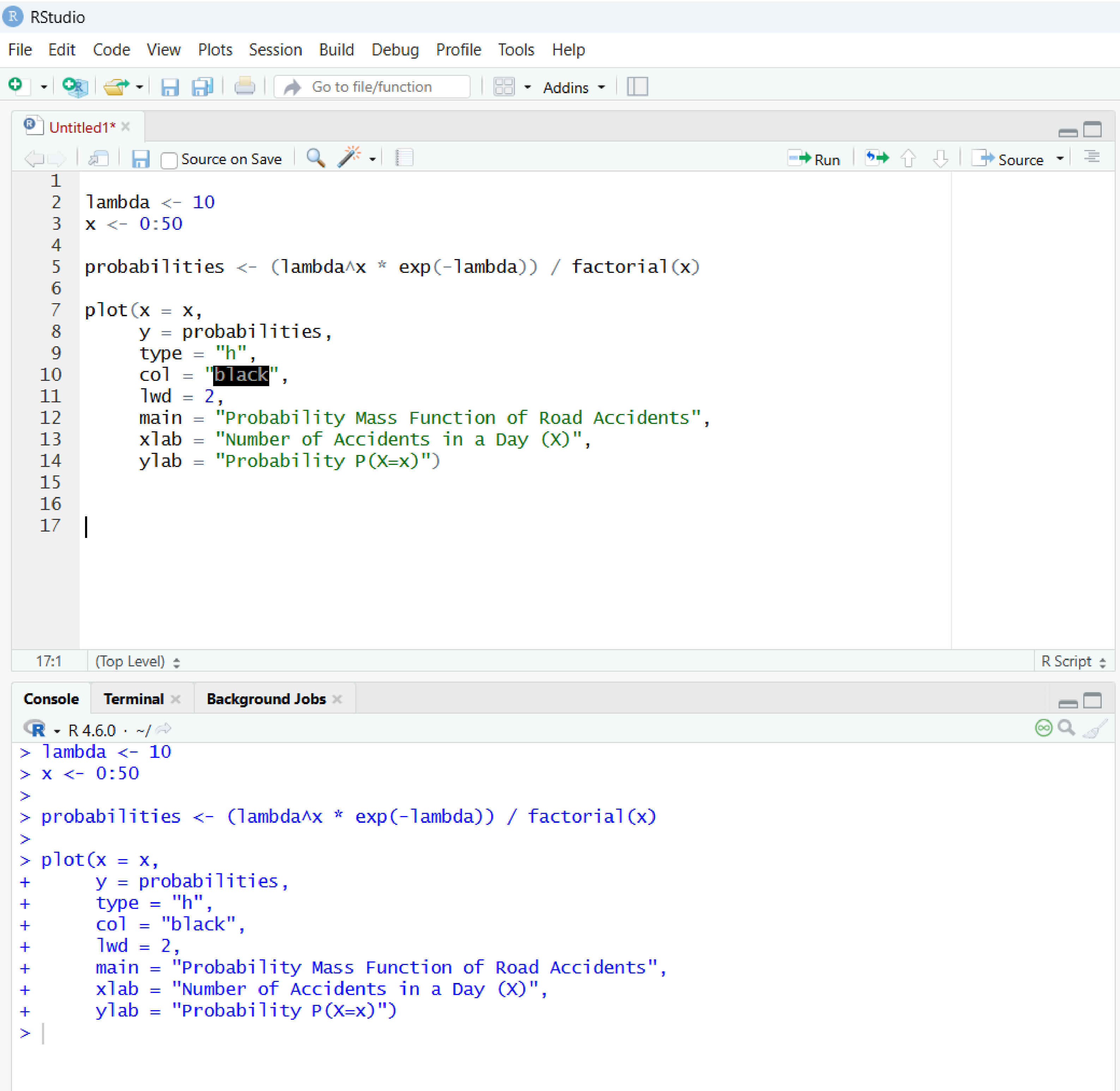
$= 0.0001126 \times 10^3$

$= 0.112599$

$f(9,10) = \frac{10^9 \times e^{-10}}{8!} = \frac{0.112599 \times 10}{9}$

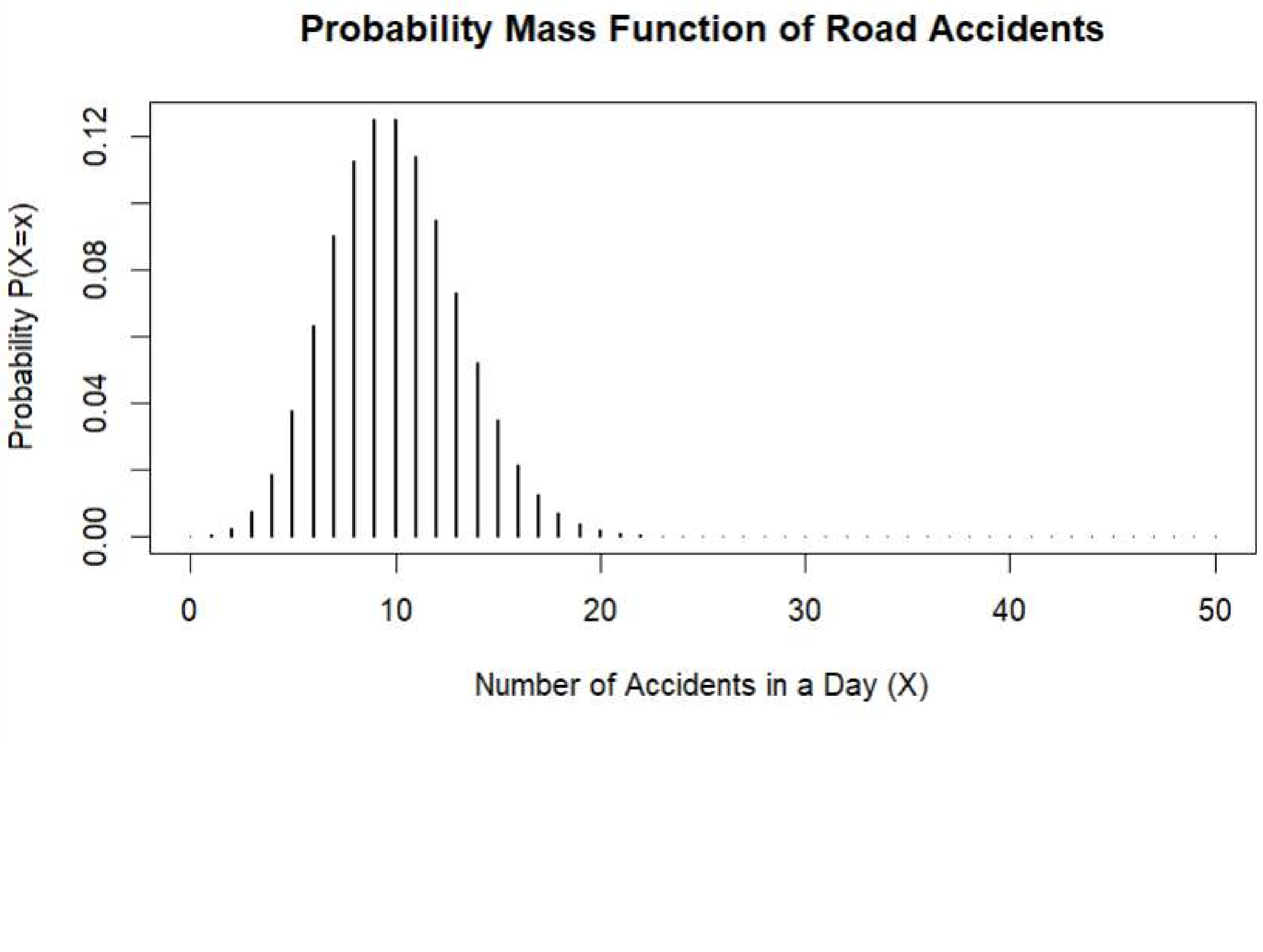
$= 0.125110$

Total probability $= f(8,10) + f(9,10) = 0.237709$


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
lambda <- 10
x <- 0:50

probabilities <- (lambda^x * exp(-lambda)) / factorial(x)

plot(x = x,
     y = probabilities,
     type = "h",
     col = "black",
     lwd = 2,
     main = "Probability Mass Function of Road Accidents",
     xlab = "Number of Accidents in a Day (X)",
     ylab = "Probability P(X=x)")
17:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> lambda <- 10
> x <- 0:50
>
> probabilities <- (lambda^x * exp(-lambda)) / factorial(x)
>
> plot(x = x,
+   y = probabilities,
+   type = "h",
+   col = "black",
+   lwd = 2,
+   main = "Probability Mass Function of Road Accidents",
+   xlab = "Number of Accidents in a Day (X)",
+   ylab = "Probability P(X=x)")
>



Probability Mass Function of Road Accidents
Probability P(X=x)
0.00
0.04
0.08
0.12
0
10
20
30
40
50
Number of Accidents in a Day (X)

Total probability $= f(8,12) + f(9,12)$

$= 0.112599 + 0.125110$

$= 0.237709$

Part 3: Continuous random variables

3.1

$X \to$ normal distribution

$$\text{Pdf} \to f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

a) $x = 0$ given $\mu = 1$, $\sigma = 1$

$$f(0) = \frac{1}{1 \times \sqrt{2\pi}} \times e^{\frac{-(0-1)^2}{2\times 1}}$$

$$= \frac{1}{\sqrt{2\pi}} \times e^{-\frac{1}{2}} = 0.3989 \times 0.6065$$

$f(0) = 0.2419$

b). $F(1) = ?$, $\mu = 0$, $\sigma = 1$

$$f(1) = \frac{1}{1 \times \sqrt{2\pi}} \times e^{\frac{-(1-0)^2}{2\times 1}} = \frac{1}{\sqrt{2\pi}} \times e^{-1/2} = 0.2419$$

c). $P(x_1 \leq X \leq x_2) = 0.3$

$P(x_1 \leq X \leq x_3) = 0.45$

$P(x_2 \leq X \leq x_3) = P(x_1 \leq X \leq x_3) - P(x_1 \leq X \leq x_2)$

$= 0.45 - 0.3$

$= 0.15$

part a : (The likelihood function)

$L(\theta|x) = f(x, \theta)$

4.1. participant 5 strings

time → 303, 443, 220, 560, 880

X = recognition time for strings.

$$X \sim f(x, \mu) = \frac{1}{x\sqrt{2\pi}} e^{\frac{-(\log x - \mu)^2}{2}}$$

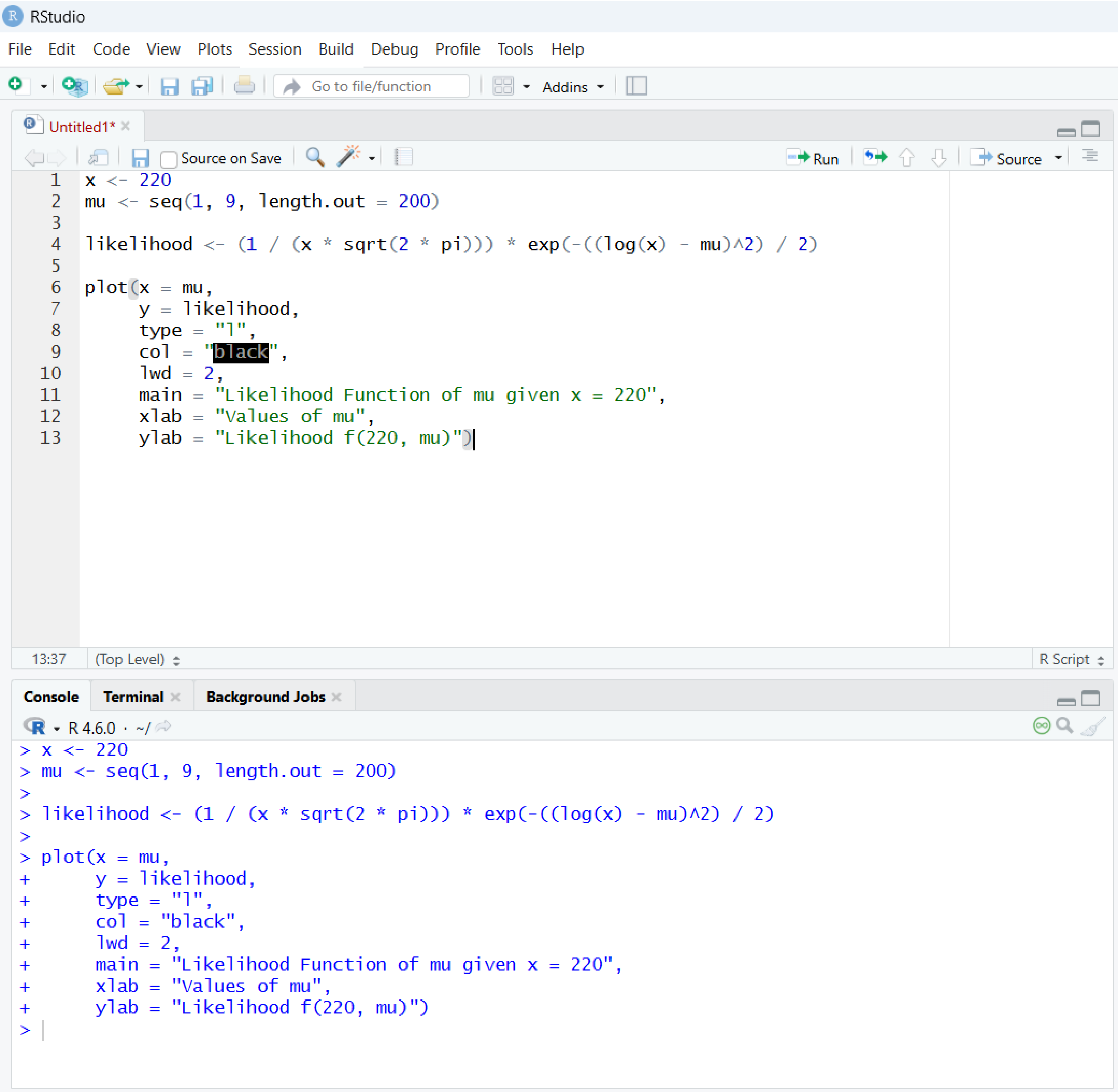
(a) $x = 220$ $\mu \longleftrightarrow$

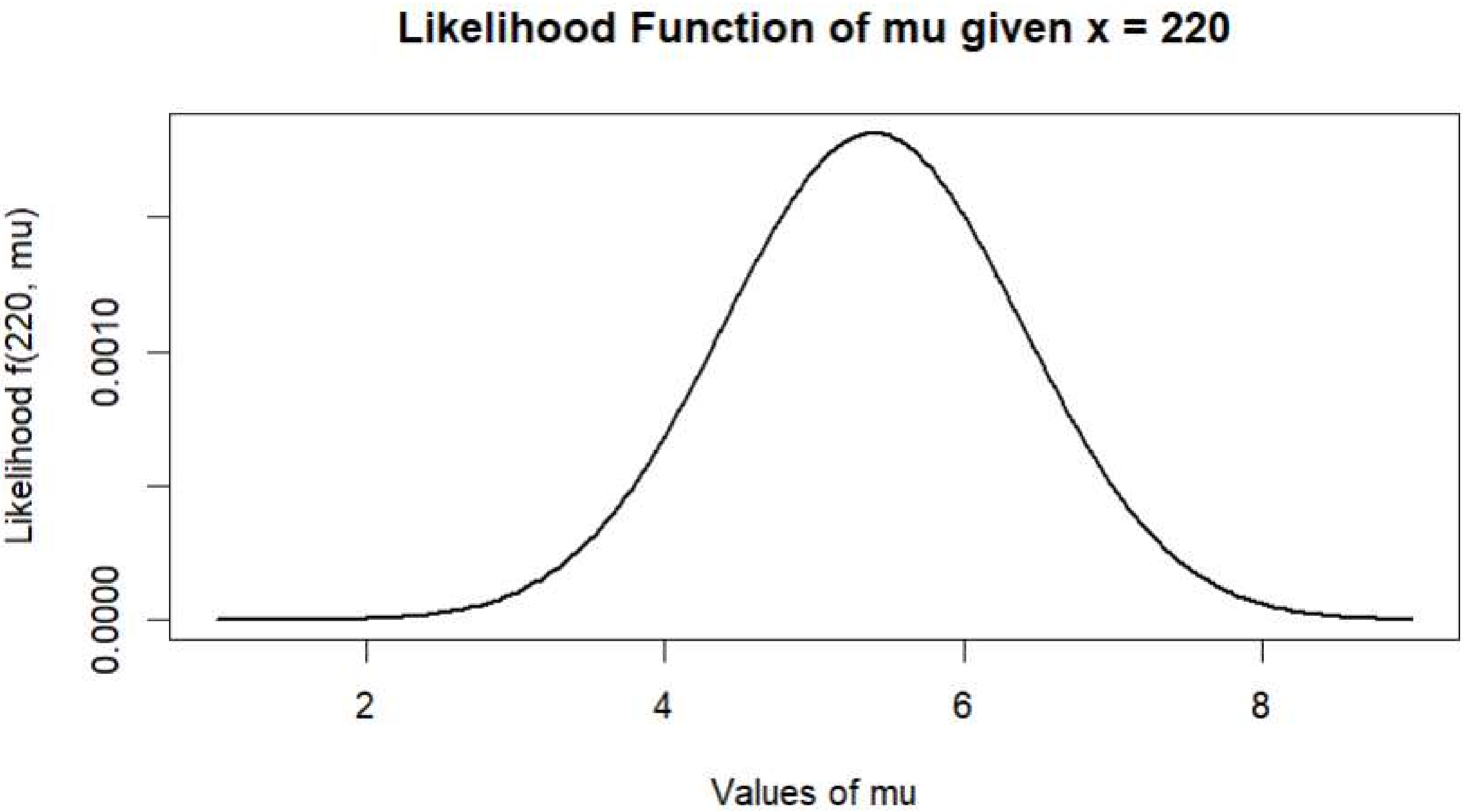
$\log 220 = 5.39$ $\mu \approx \underline{514} \rightarrow \underline{1409}$

b)

c) The maximum value is around 6.05 - 6.15 so the value can be around ≈ 6.075

(from graph if we calculate we are getting exact value 6.061765)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
x <- 220
mu <- seq(1, 9, length.out = 200)
likelihood <- (1 / (x * sqrt(2 * pi))) * exp(-((log(x) - mu)^2) / 2)
plot(x = mu,
y = likelihood,
type = "l",
col = "black",
lwd = 2,
main = "Likelihood Function of mu given x = 220",
xlab = "Values of mu",
ylab = "Likelihood f(220, mu)")
13:37
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> x <- 220
> mu <- seq(1, 9, length.out = 200)
>
> likelihood <- (1 / (x * sqrt(2 * pi))) * exp(-((log(x) - mu)^2) / 2)
>
> plot(x = mu,
+ y = likelihood,
+ type = "l",
+ col = "black",
+ lwd = 2,
+ main = "Likelihood Function of mu given x = 220",
+ xlab = "Values of mu",
+ ylab = "Likelihood f(220, mu)")
>



Likelihood Function of mu given x = 220
Likelihood f(220, mu)
0.0000
0.0010
2
4
6
8
Values of mu

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Untitled1*

Source on Save Run Source

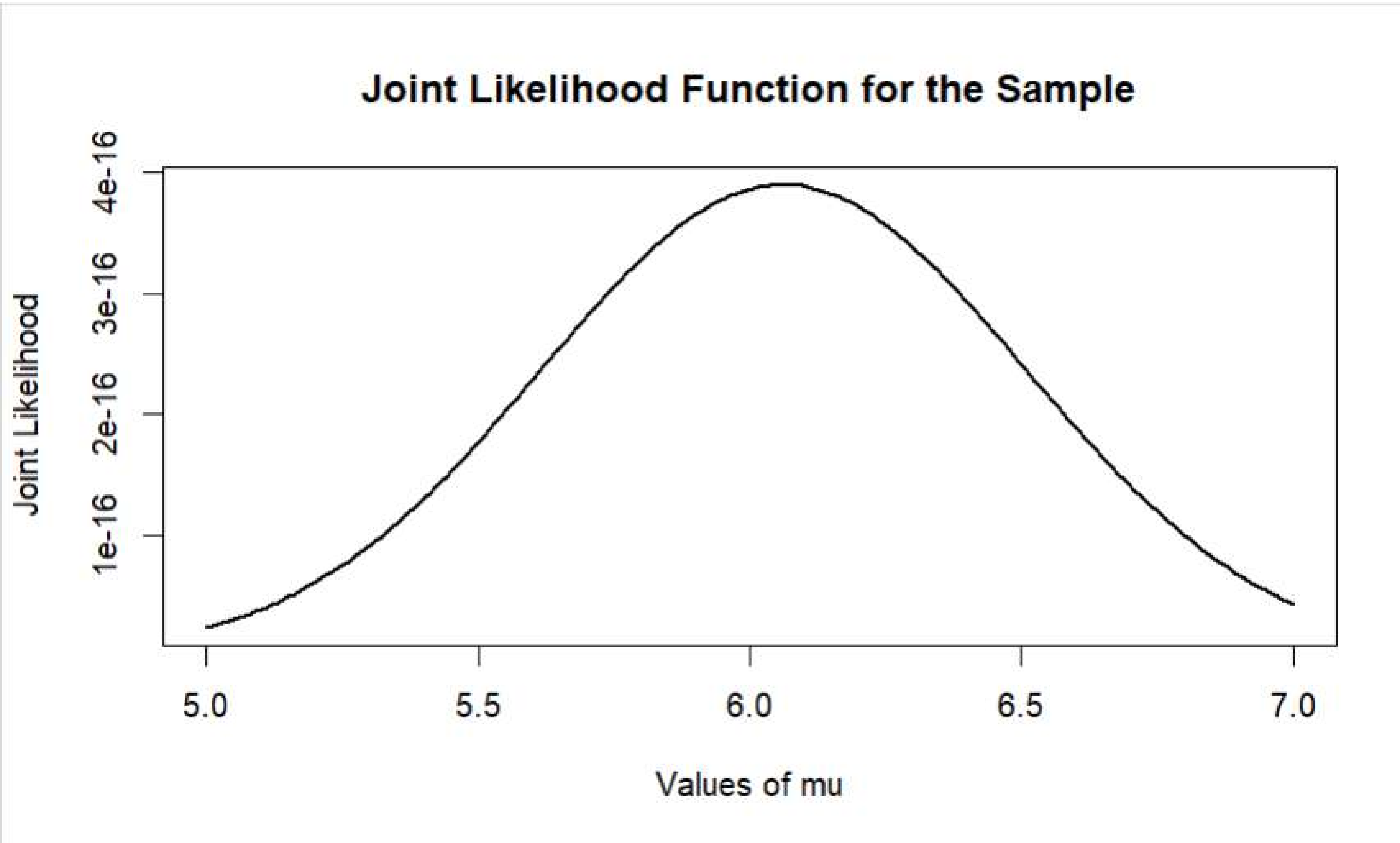
```r
x <- c(303.25, 443, 220, 560, 880)
n <- length(x)
inv_prod <- 1 / prod(x)

mu <- seq(5, 7, length.out = 200)
likelihood <- numeric(length(mu))

for (i in 1:length(mu)) {

  sumofsqrs <- sum((log(x) - mu[i])^2)

  likelihood[i] <- inv_prod * (1 / sqrt(2 * pi))^n * exp(-sumofsqrs / 2)
}

plot(x = mu,
     y = likelihood,
     type = "l",
     col = "black",
     lwd = 2,
     main = "Joint Likelihood Function for the Sample",
     xlab = "Values of mu",
     ylab = "Joint Likelihood")
```

17:18 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.0 · ~/

```
> x <- c(303.25, 443, 220, 560, 880)
> n <- length(x)
> inv_prod <- 1 / prod(x)
>
> mu <- seq(5, 7, length.out = 200)
> likelihood <- numeric(length(mu))
>
> for (i in 1:length(mu)) {
+
+   sumofsqrs <- sum((log(x) - mu[i])^2)
+
+   likelihood[i] <- inv_prod * (1 / sqrt(2 * pi))^n * exp(-sumofsqrs / 2)
+ }
>
> plot(x = mu,
+      y = likelihood,
```


Joint Likelihood Function for the Sample
Joint Likelihood
1e-16
2e-16
3e-16
4e-16
5.0
5.5
6.0
6.5
7.0
Values of mu